ALHASSAN JAWAD

ELECTRICAL ENGINEER • FOCUS: CYBER- & OT-SECURITY

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PROFESSIONAL SUMMARY

Electrical Engineer (B.Sc.) currently working part-time at Abbott Laboratories while simultaneously completing a M.Sc. in Electrical Engineering (expected Q1 2027) with coursework centered on **OT-security** and **control of electrical systems**.

My background bridges **electrical infrastructure** & **control theory** with **programming**, applying **automation logic** to embedded architectures. Focus is directed toward engineering secure, resilient control systems for critical infrastructure.

EDUCATION

Master of Science (M.Sc.) in Electrical Engineering 2025 – 2027
Uppsala University – Specialization: Control of Electrical Systems

Bachelor of Science (B.Sc.) in Electrical Engineering 2019 – 2022
Department of Electrical Engineering | Uppsala University
Degree issued in 2026

TECHNICAL COMPETENCIES

Programming Languages: Python, C/C++, MATLAB, LabVIEW

Engineering Tools & Frameworks: Git, Altium Designer, PSpice, LTSpice, NIST SP 800-82, ISO/IEC 27001

Engineering Skills: Automation, Control Systems, Signal Processing, Data Analysis

Languages: Swedish (Fluent), English (Fluent), Arabic (Native)

CERTIFICATIONS

- **micro1:** Vetted Software Engineer (Electronic Engineering / Ngspice) [Issued 2026]
- **Google Cloud Certified**: Professional Google AI Engineer [Issued 2026]

RELEVANT EXPERIENCE

IT Infrastructure Specialist - Abbott Laboratories Sep 2026 - Ongoing

Provide on-site infrastructure maintenance and client support for Abbott's BTS facility in Solna, serving as the Tier-3 escalation point for divisions across Sweden, Denmark, and Finland. Configure and deploy enterprise hardware, troubleshoot system images, and resolve physical hardware faults directly on site. Administer Active Directory user provisioning, manage access controls, and maintain local network connectivity across site departments.

Software Engineer (Internship) – Portabel Health Sep 2025 – Dec 2025

- Maintained backend data queries and schema adjustments using PostgreSQL.
- Authored system refactoring documentation and established internal Git branch management workflows to ensure code traceability.
- Supported technical testing and verification across development and QA environments.

Teaching Assistant – Uppsala University Aug 2023 – Jul 2025

- Instructed laboratory sessions for 50+ engineering students per semester in fundamental electrical engineering, circuit theory, and programming.
- Supervised lab safety, assisted students with hardware debugging using oscilloscopes and multimeters, and graded technical lab reports and exams.
- Collaborated with department faculty to update laboratory guidelines and verify measurement setups.

SELECTED PROJECTS

Master's Thesis: Automated Aeroponic Cultivation System
2025

Internal Collaboration with Uppsala University

- Programmed and deployed a multi-variable MIMO rule-based feedback control loop on an ESP32 microcontroller to regulate closed-loop temperature and relative humidity.
- Integrated an array of digital environmental sensors over an I2C telemetry network to establish a continuous data acquisition loop for system state verification.
- Compiled technical documentation covering system initialization, register configurations, and multi-variable tuning routines.

Bachelor's Thesis: Smart Building Energy Management
2022

External Collaboration with Studenternas IP, Uppsala

- Wrote a custom optimization script in MATLAB to manage energy consumption and load distribution for a commercial property.
- Processed multi-variable datasets from facility power meters, isolating 71.2 kW out of a 185 kW peak load as manageable power flexibility.
- Proved that 38.5% of the infrastructure's peak demand could be shifted to reduce peak grid tariffs without affecting baseline operations.

Project: Miniature Elevator Prototype (Course: Project in Electronics 10 ECTS)
2023

Academic Project | Uppsala University

- Designed a three-level miniature elevator, handling end-to-end HIL (Hardware-In-the-Loop) testing.
- Programmed the control logic and GUI (in LabVIEW), to manage real-time level selection and state transitions while regulating motor speed profiles, to achieve a smooth acceleration while minimizing physical oscillations.

Project: GPS Tracking Device (Course: PC-Board Construction with ECAD Tools 10 ECTS)
2022

Academic Project | Uppsala University

- Designed a multi-layer custom PCB in Altium Designer adhering to component isolation constraints.
- Integrated an ATmega328P microcontroller with a Quectel MC60 transceiver to process satellite GPS data and transmit coordinates over different networks.
- Implemented a hardware power-saving scheme by interfacing an MMA8653 accelerometer to provide system wake-up triggers based on physical motion.

References

Available upon request

Thank you for taking the time to look at my CV
If you have any questions, don't hesitate to contact me @ alhassanjawad.com/#contact